National Booksellers Inc.

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Team 8 ~~ Yiyi Su, Xingjian Murong, Brealin Redecker

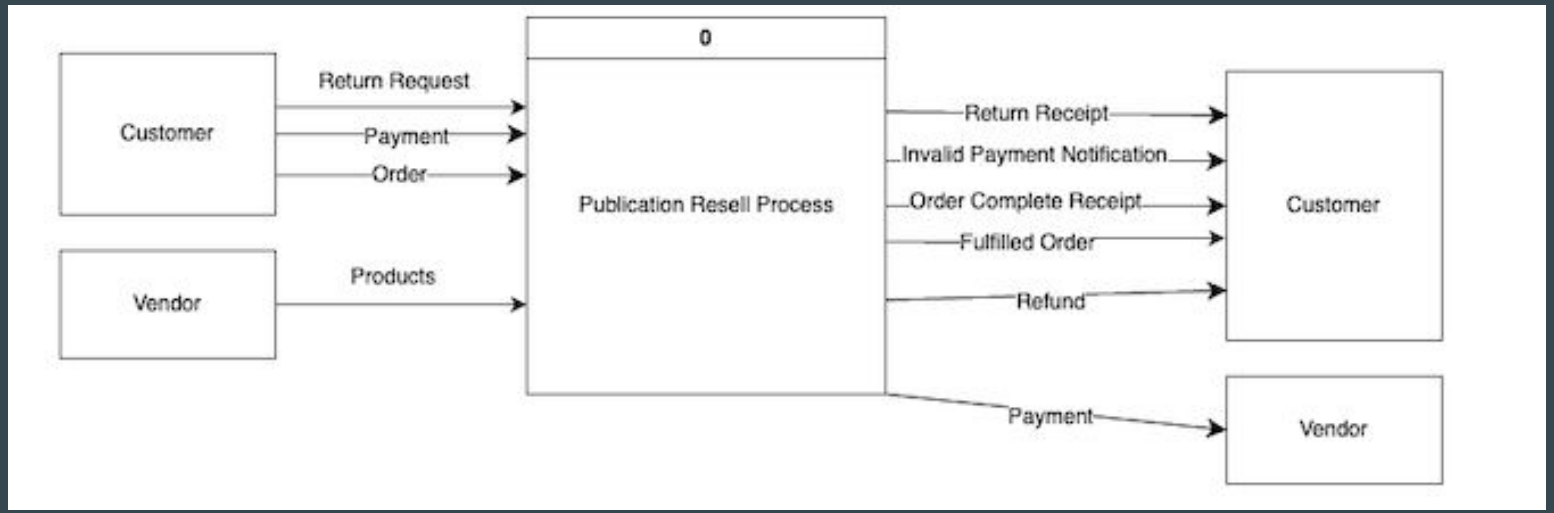
NBI Overview

- **General Gist:**
  - Book and Compact Disc Resellers
- **Main Operations:**
  - Customers, Order Entry, Publishers/Vendors, Inventory, Invoicing, Shipping, Returns
- **Issues:**
  - Limited Support with Vendors
  - Infrequent Updates
  - Inadequate Performance for # of Users
  - Slow and Unstable
  - Lacks adequate reporting features

Scope & System Requirements

- **Scope:**
  - The NBI Information System will process orders, process purchasing, process returns, update inventory received from Vendors.
- **Functional Requirements:**
  - Process customer orders.
  - Handle products purchasing from vendors/publishers.
  - Process customer orders return requests with refund.
  - Store customer and publisher/vendor information.
- **Non-Functional Requirements**
  - Support the communication for at least 100 users across both locations.
  - Support documents printing.
  - Support online and business phones for customer services.
  - Restrict system access based on user roles.
  - Provide customer purchasing statements every 30 days.

DFD Analysis (Context)

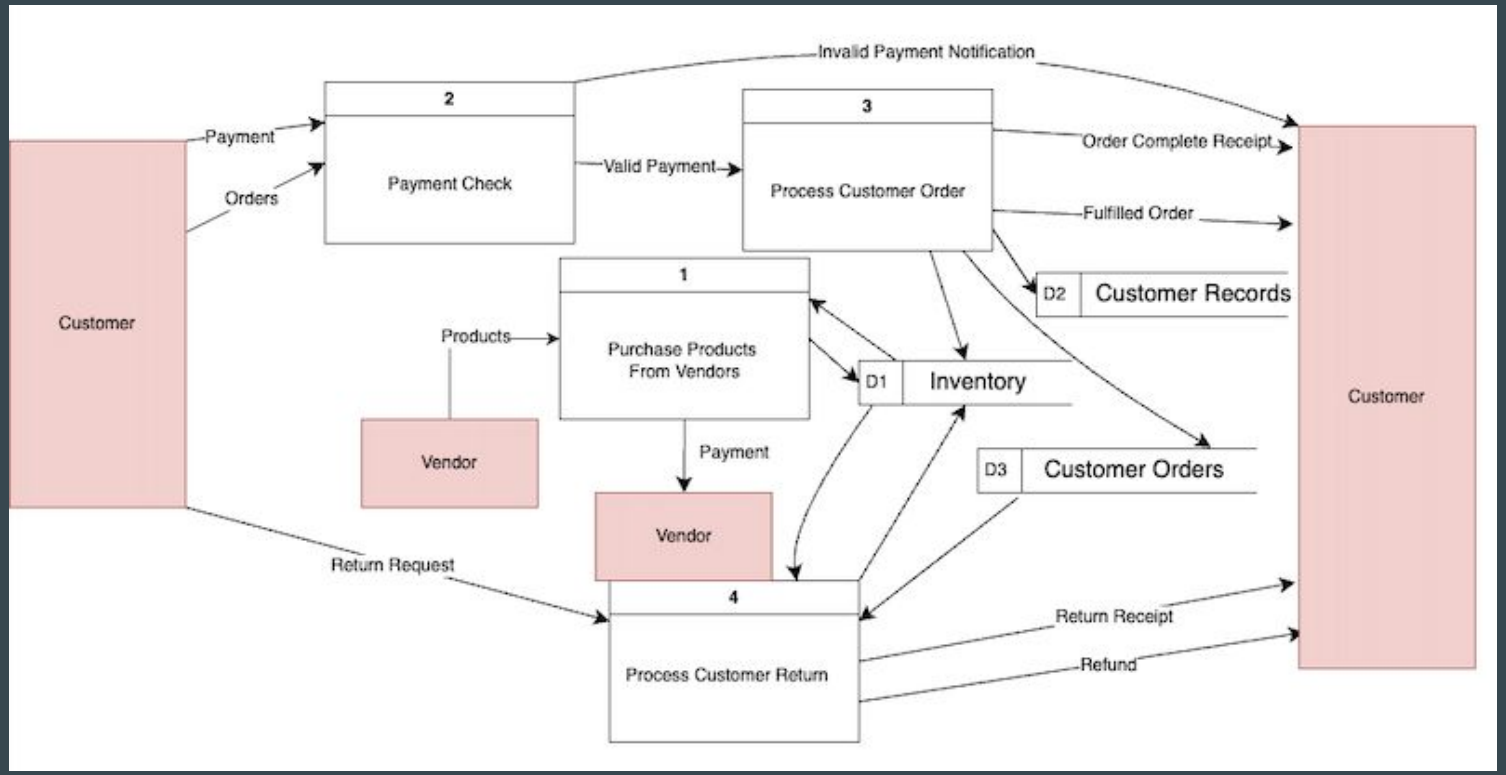


External Entities: Customer, Vendor

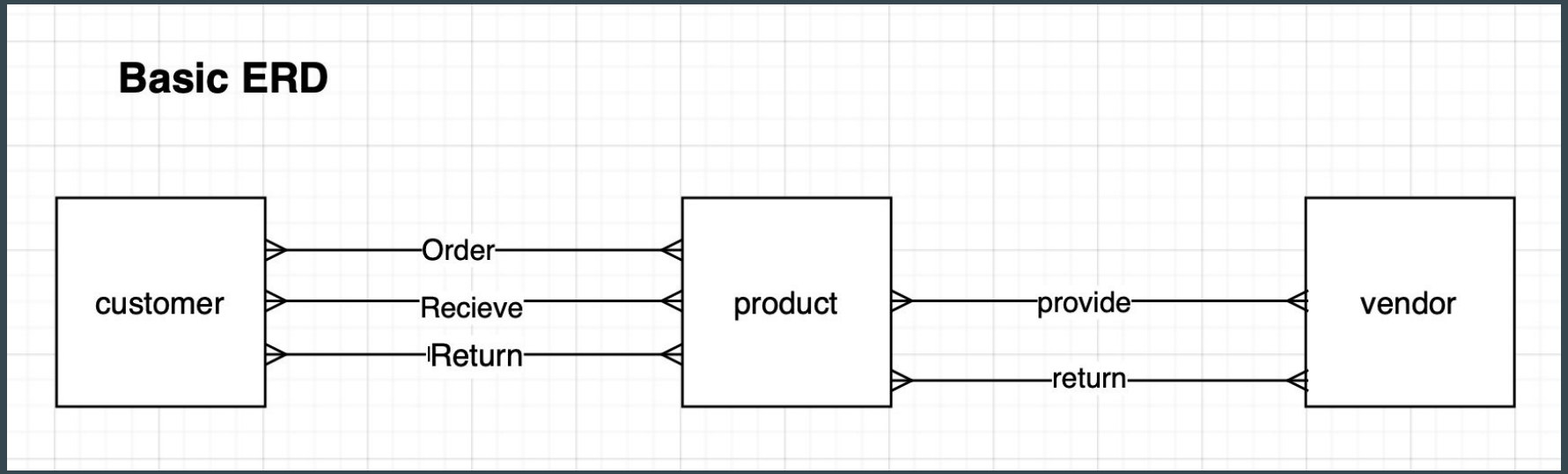
DFD Analysis (Business Assumptions)

- A receipt will be sent to the customer for each order/return request; a failure notification will be sent to the customer if the payment is invalid.
- Each transaction will update the inventory.
- A new customer record will be made when processing each order, and the order item details will be stored into the customer order datastore. If there is a repeating customer, new details will be updated in the customer order datastore.
- The information in the customer order datastore will be referenced when processing return requests.
- A refund will be sent to the customer after processing the return request.

DFD Analysis (Level 0)

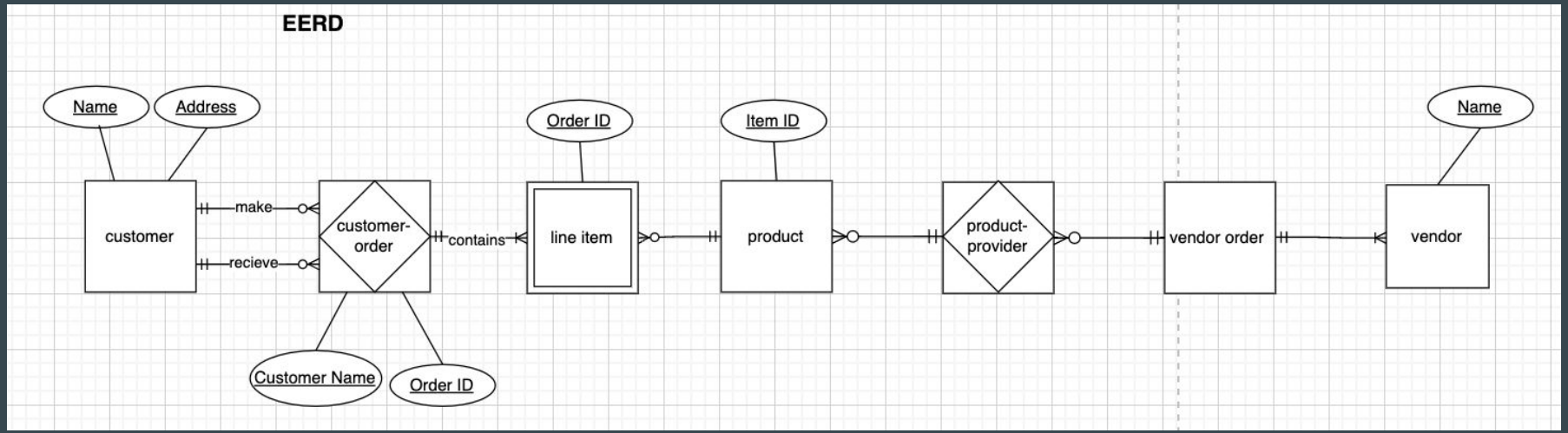


ERD Analysis

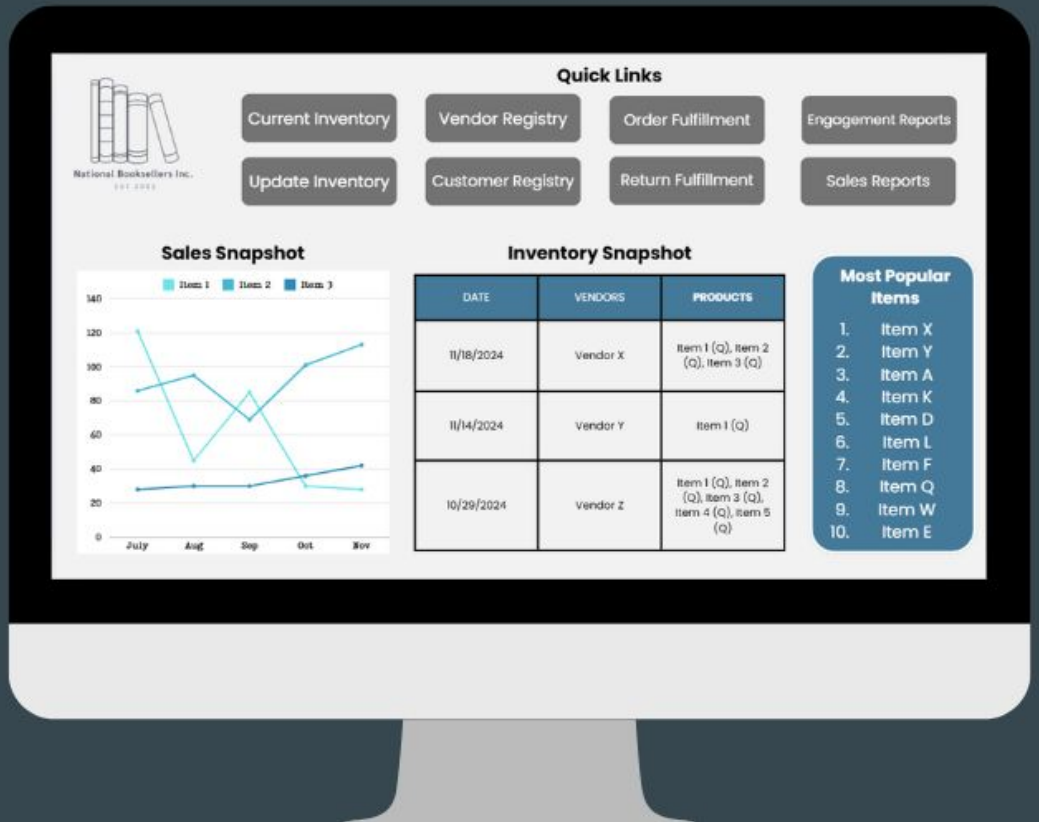


Fundamental Entities: Customer, Product, Vendor

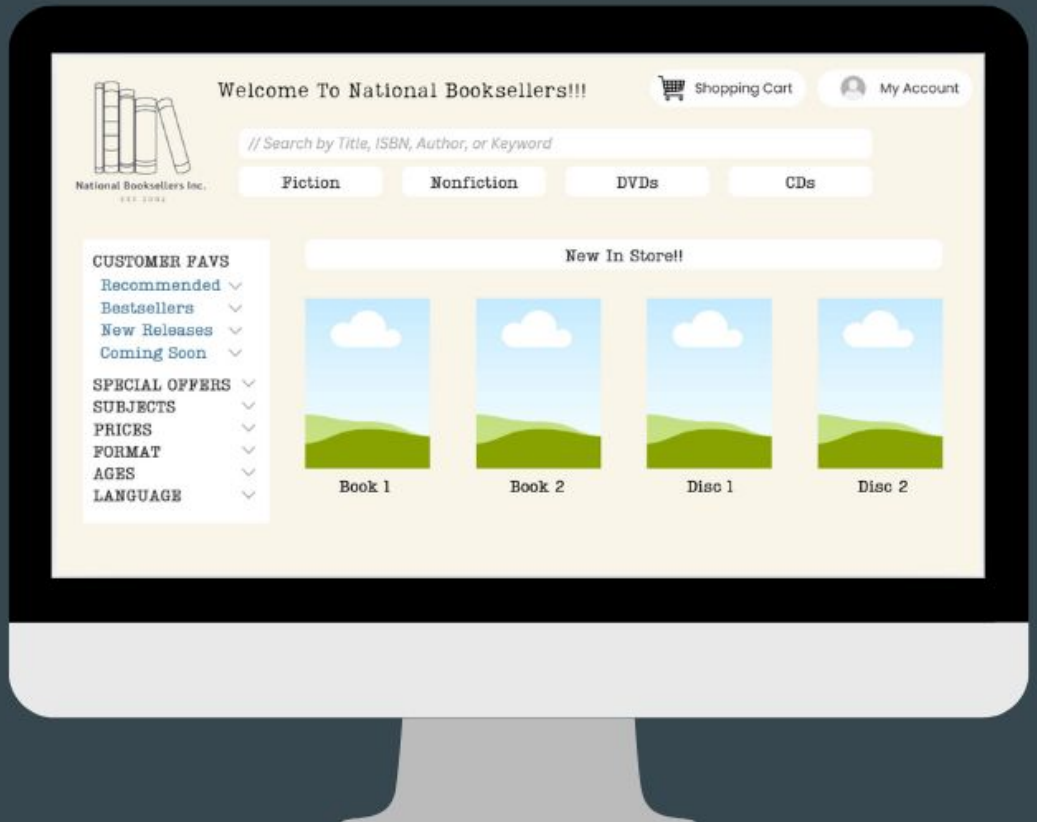
EERD Analysis



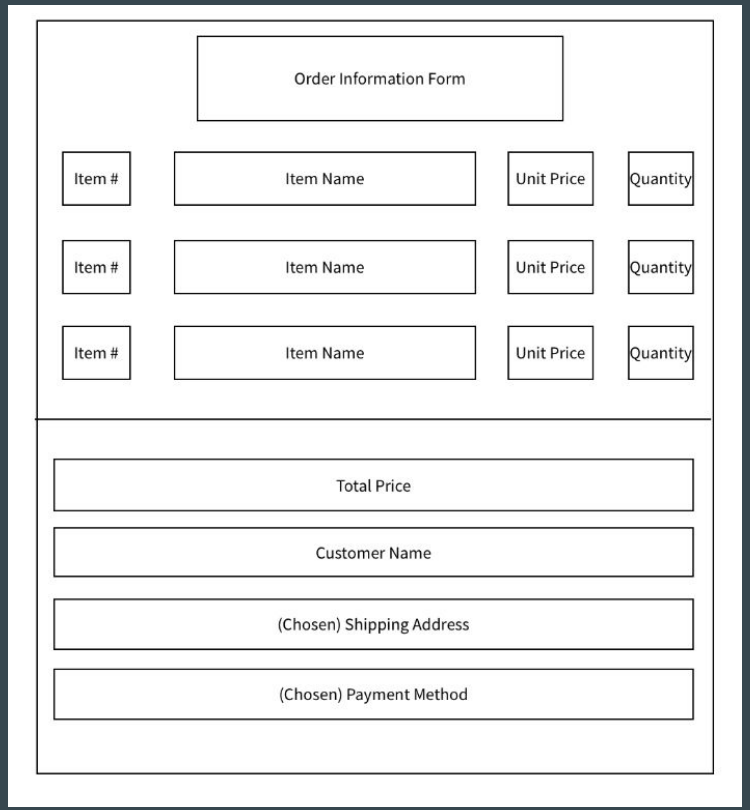
Interface (Manager)

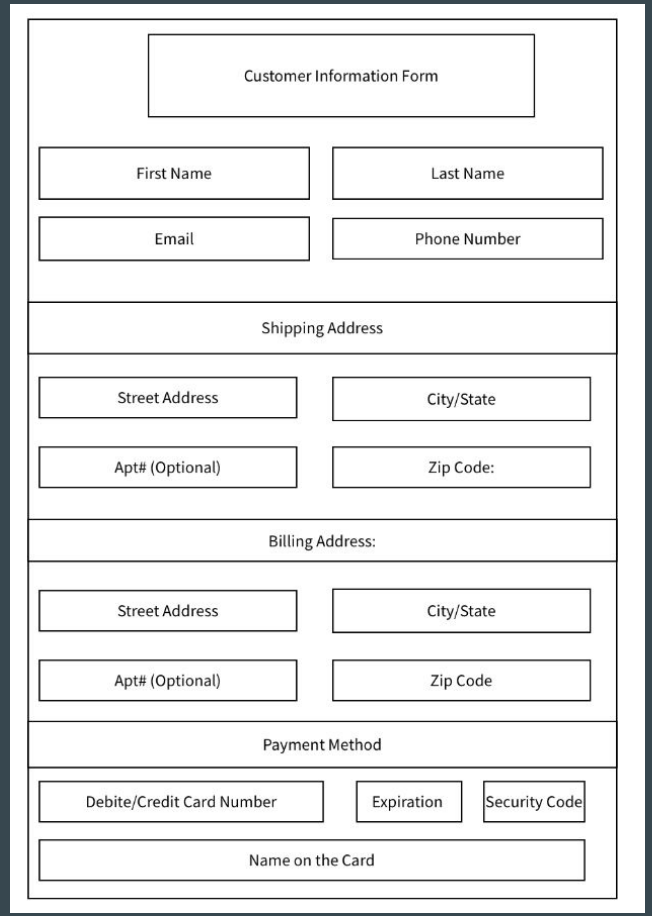


Interface (Customer)



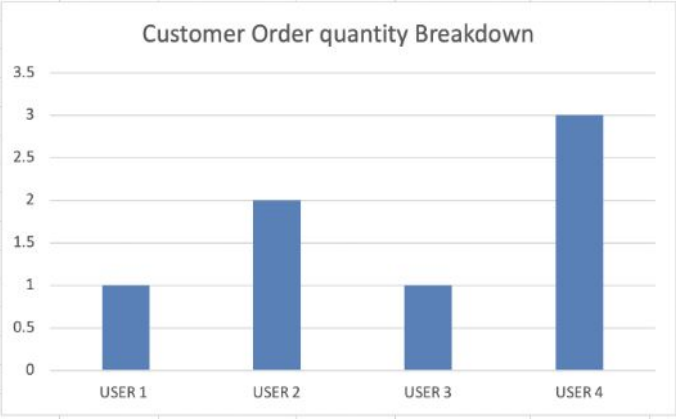
Input Forms


Order Information Form
Item #
Item Name
Unit Price
Quantity
Item #
Item Name
Unit Price
Quantity
Item #
Item Name
Unit Price
Quantity
Total Price
Customer Name
(Chosen) Shipping Address
(Chosen) Payment Method


Customer Information Form
First Name
Last Name
Email
Phone Number
Shipping Address
Street Address
City/State
Apt# (Optional)
Zip Code:
Billing Address:
Street Address
City/State
Apt# (Optional)
Zip Code
Payment Method
Debite/Credit Card Number
Expiration
Security Code
Name on the Card

Output Reports

A	B	C	D	E	F	G
Customer N	Order ID	Product Name	Quantity	Order Date	Order Status	Refund Amount
USER 1	ORD001	Book A	1	2024-11-01	Delivered	-
USER 2	ORD002	Book B	2	2024-11-02	Pending	-
USER 3	ORD003	Book C	1	2024-11-03	Returned	$15
USER 4	ORD004	Book D	3	2024-11-04	Delivered	-



A	B	C	D	E	F
Vendor Name	Vendor Order ID	Product Name	Quantity Received	Transaction Date	Status
vendor 1	VEND001	Book A	100	2024-10-20	Completed
vendor 2	VEND002	Book B	50	2024-10-21	Pending
vendor 3	VEND003	Book C	75	2024-10-22	Completed
vendor 4	VEND004	Book D	30	2024-10-23	Returned

Frustrations and Difficulties

Yiyi

It is very time consuming to include everything.

Brealin

Repetition and experience speaking the languages that are being a systems analyst.

Xingjian

Balancing diagram details and separating requirements was challenging.

Experience and Lessons

Yiyi

Understanding the requirements is the first step to successfully building a system.

Brealin

You have to think big picture, but also be honed in on the small details.

Xingjian

Feedback and teamwork improved our process.

Question?